Solution Requirements

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	*No login required*	Low
2	Authorization levels	*Public access*	Low
3	External interfaces	*Accept user inputs through web form*	High

4	Transactions processing	*Process input values and predict Flood Risk*	High
5	Reporting	*Display Flood Risk score and category*	High
6	Business rules, etc.	*Predict using XGBoost Classifier model*	High
7	Compliance to laws or regulations	*Educational purpose only*	Medium
8	*Other*	*Data visualization support*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	*Prediction should be fast*	*<2 seconds*
2	Scalability	*Can handle multiple predictions*	*100+ requests*
3	Security & Data Privacy	*Validate numeric inputs*	*Safe input handling*
4	Reliability & Availability	*Stable Flask application*	*99% availability*
5	Usability & Accessibility	*Simple responsive UI*	*Easy navigation*
6	*Other*	*Maintainable code*	*Modular Python files*